Module 2 Assignment

1. Disease & data choice

Ovarian cancer is a heterogeneous group of malignancies with distinct histologic and molecular subtypes. High-grade serous ovarian cancer (HGSOC) is the dominant subtype and is responsible for a large proportion of ovarian cancer deaths. HGSOC is also characterized by extensive genomic instability and frequent alterations in pathways involved in DNA repair, contributing to its aggressive clinical behavior and making it particularly relevant for multi-omics target discovery (Lisio et al., 2019). These molecular features make ovarian cancer a suitable disease for multi-omics analysis, because integrating transcriptomic, proteomic, and genomic evidence may help identify genes supported across multiple biological layers.

For the transcriptomic layer, I first considered the CPTAC ovarian cancer RNA-seq data from LinkedOmics. However, only tumor RNA data were available, so tumor and normal effect sizes and p-values could not be calculated. So I used Expression Atlas experiment E-GEOD-40595, which provides gene-level log2 fold-change and p-value results for ovarian cancer and normal epithelial cells. The dataset was openly accessible (Papatheodorou et al., 2018).

For the proteomic layer, I used gene-level CPTAC ovarian cancer proteomics data from LinkedOmics. Tumor and normal protein abundance were compared for each gene to obtain a protein effect and p-value. The dataset was openly accessible (Vasaikar et al., 2018).

For the genomic layer, I used gene-level somatic mutation data from the same CPTAC ovarian cancer cohort through LinkedOmics. Mutation frequency was calculated for each

gene and used as the genomic association-strength measure. The dataset was also openly accessible (Vasaikar et al., 2018).

The transcriptomic data and the CPTAC proteomic/genomic data came from different cohorts, so they were not matched at the patient level. Therefore, the three omics layers were integrated at the gene level. A total of 4,396 genes survived the three-way join. Among them, there are 2,120 showing concordant RNA and protein changes, while 2,276 were classified as non-concordant.The final multi-group comprehensive score is calculated by the weight of the three group layers.

2. Top targets

RUNX1T1 was the highest-ranked gene in the multi-omics analysis. Its multi-omics score is 0.9796, RNA expression decreases (−4.2), and protein abundance also decreases (−1.87). Open Targets showed existing associations with ovarian cancer-related disease terms, providing additional support for its biological relevance. This made RUNX1T1 a useful example of a top-ranked gene that is also connected to known cancer biology(Buniello et al., 2025). PGR ranks second, with a multi-group score of 0.9772. RNA expression and protein abundance have decreased.

GREB1 also showed ovarian cancer-related evidence in Open Targets, including literature and expression-related evidence. This supports GREB1 as a biologically plausible hit recovered by the multi-omics ranking(Buniello et al., 2025). Also, The RNA expression of GREB1 decreased (−2.6), but the protein abundance increased (+0.90). This shows that there is discordance between the two layers of RNA and protein.

COL3A1 showed some ovarian cancer-related association in Open Targets, but the evidence appeared less established than for the other two genes. For this reason, COL3A1 may be

considered a less obvious candidate that could be worth further investigation.(Buniello et al., 2025)

Together, these results suggest that the ranking recovered both biologically supported ovarian cancer-related genes and less obvious candidates that may merit additional study.

3. RNA–Protein Discordance

A2ML1 showed discordant RNA and protein changes in this analysis, with increased RNA expression but decreased protein abundance. The RNA log2 fold change of A2ML1 is +0.8, while the protein effect value is −1.48.

RNA and protein levels do not always change in the same direction. In ovarian cancer cells, previous proteomic work has shown that protein and mRNA expression changes can be discordant, possibly because of post-transcriptional regulation (Stewart et al., 2006). Therefore, the opposite RNA and protein changes observed for A2ML1 may reflect regulation after transcription, although the exact mechanism for A2ML1 is not established. In addition, the transcriptomic and proteomic data came from different cohorts, so differences between patient groups may also contribute to the disagreement. Therefore, this result should be interpreted as a gene-level observation rather than a confirmed biological mechanism.

4. Limitations and Interpretation

Since the three datasets were combined by gene rather than by the same patients, this analysis can show which genes have evidence across multiple omics layers. These genes can be considered possible candidates for further study. However, the RNA, protein, and mutation changes may not have happened in the same tumors. Also, a high multi-omics score does not mean that a gene directly causes ovarian cancer or that it will be a good drug target. More experiments would be needed to confirm these findings. Cancer driver genes cannot just be determined by mutation frequency alone, and multi-group scoring cannot directly measure

the druggability. In the future, the biological role of priority candidate genes can be verified by independent ovarian cancer samples and further experiments.

Reference

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